

Spatio-Temporal Predictive Learning Using Crossover Attention for Communications and Networking Applications

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ABSTRACT This paper investigates the spatio-temporal predictive learning problem, which is crucial in diverse applications such as MIMO channel prediction, mobile traffic analysis, and network slicing. To address this problem, the attention mechanism has been adopted by many existing models to predict future outputs. However, most of these models use a single-domain attention which captures input dependency structures only in the temporal domain. This limitation reduces their prediction accuracy in spatio-temporal predictive learning, where understanding both spatial and temporal dependencies is essential. To tackle this issue and enhance the prediction performance, we propose a novel crossover attention mechanism in this paper. The crossover attention can be understood as a learnable regression kernel which prioritizes the input sequence with both spatial and temporal similarities and extracts relevant information for generating the output of future time slots. Simulation results and ablation studies based on synthetic and realistic datasets show that the proposed crossover attention achieves considerable prediction accuracy improvement compared to the conventional attention layers.

INDEX TERMS Spatio-temporal, multivariate time series, traffic prediction, crossover attention, transformer model, deep learning.

I. INTRODUCTION

SPATIO-TEMPORAL multivariate time series (STMTS) are defined by their sequential order and spatio-temporal dependencies, containing valuable information into the dynamics of various systems and processes in communications and networking. Spatio-temporal predictive learning seeks to generate future frames of STMTS by analyzing the available historical frames. The importance of spatio-temporal predictive learning lies mainly in its ability to analyze and model both the spatial correlations and temporal state transitions of the system dynamics. Analyzing and modeling STMTS is a crucial aspect of data mining, providing essential insights and informing decisions across various applications such as multiple-input multiple-output (MIMO)

channel prediction [1], [2], [3], mobile traffic analysis [4], [5], network slicing [6], [7], and smart cities [8]. For instance, accurate and timely mobile traffic prediction is needed for the intelligent resource management in network slicing [9], which could mitigate network congestion and improve the quality of services (QoS). Meanwhile, channel prediction can help solve the channel aging issue [1], reduce the pilot overhead [3] and thereby enhance the system performance.

In spatio-temporal predictive learning, understanding both the temporal and spatial dependencies is crucial for enhancing prediction accuracy. Conventional statistical techniques like Historical Average (HA) and Auto-Regressive Integrated Moving Average (ARIMA) often perform poorly in this task, as they were designed to capture only temporal

correlations [10]. Additionally, these traditional algorithms inefficiently deal with the non-linear dynamics of time series data. Fortunately, the remarkable advancements in deep learning over the past decade have led researchers to explore numerous data-driven approaches to tackle this challenge, enabling adaptive learning and the modeling of complex non-linear dependencies in spatio-temporal predictive learning.

A. LITERATURE REVIEW

Initially, pioneering studies in communication and networking were conducted by exploring Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) to predict STMTS [2], [11], [12]. In practice, researchers frequently integrate RNNs and CNNs as a hybrid network to simultaneously learn patterns in data over time and space domain [13]. This is because RNNs are good at capturing temporal dependencies, while CNNs focus on identifying spatial dependencies. It has been shown that such hybrid architectures can efficiently capture the spatio-temporal dynamics in time series forecasting. For example, the authors in [14] demonstrated that a neural network with convolutional long-short term memory (ConvLSTM) layers significantly outperform conventional linear regression methods in the spatio-temporal MIMO channel prediction problem. In [15], the authors leveraged CNN and ConvLSTM to predict channel state information (CSI) in high-speed railway networks and verified that the hybrid architecture outperformed the classical CNN and RNN networks in terms of prediction accuracy.

Besides CNNs, Graph Convolutional Networks (GCNs) have recently attracted significant research interest in spatio-temporal forecasting. Unlike CNNs, GCNs are specifically designed to better capture the spatial correlations of graphs in non-Euclidean space. As a result, GCNs are considered as better substitutions for the CNN modules in existing networks to efficiently understand the spatial structure of data. For example, the authors of [16] proposed to model mobile terminals as a time-evolving graph and used GCN to predict the future mobile traffic data. Their experiments demonstrated that the proposed GCN-based model outperformed CNN-based models across various prediction metrics. Likewise, the authors of [17] demonstrated that a GCN backbone achieves superior traffic flow prediction performance compared to CNN backbones.

RNN models, such as LSTM, are widely recognized for their limitations in handling long-range temporal dependencies [18]. To address this issue, Transformers based on attention mechanisms have recently been proposed [7]. One of the key advantages of Transformers is their ability to capture long-range dependencies and interactions [19]. This capability, stemming from the core attention mechanism within Transformers, is particularly beneficial for time series modeling, leading to significant advancements across various applications. For example, the authors in [20]

proposed spatio-temporal Transformer networks (STTN) for traffic flow prediction, demonstrating significant improvements in prediction accuracy compared to ConvLSTM [21]. The authors in [6] employed the attention mechanism in spatio-temporal cellular mobile traffic prediction, achieving a higher prediction accuracy compared with RNN backbones. The encoder-decoder Transformer has shown its advantages in MIMO channel prediction [22], which can accurately predict future channels in parallel. In [23], the authors employed a hybrid network including GCNs and attention mechanism to predict cellular traffic. It has been shown therein that the hybrid architecture effectively leverage the temporal dependencies of cellular traffic and spatial dependencies of the physical network topology.

B. MOTIVATIONS AND CONTRIBUTIONS

The continuous development of spatio-temporal predictive learning for time series has highlighted the essential role of Transformers for the temporal module of hybrid network architectures [24], [25]. The attention model, which forms the foundation of Transformer models, has recently been extensively investigated in time series forecasting [18]. Different variants such as Pyraformer [26], Informer [27], and Reformer [28] primarily focus on reducing computational complexity without compromising prediction performance. Consequently, the vanilla attention model remains widely used in many applications [6], [7], [21], [22], [23], [29], [30], provided the sequence length is manageable. Despite its common use for handling spatially and temporally correlated data, the vanilla attention model was originally designed as a learnable regression kernel based solely on temporal similarities [24], which potentially limits its capability for spatio-temporal predictive learning. For this reason, the existing approaches often integrate Transformers models and convolutional networks to jointly capture the spatio-temporal correlations among data frames [7], [22], [31]. Based on this consideration, we argue that addressing this issue can further enhance its efficiency in predicting spatially and temporally correlated data.

In this work, we propose a novel *Crossover Attention (XOA)* model for spatio-temporal predictive learning. It functions as a learnable regression kernel that predicts values by simultaneously considering both spatial and temporal similarities. This feature is particularly appealing for spatio-temporal predictive learning, as it prioritizes input sequences with both spatial and temporal similarities, extracting relevant information for generating future outputs. Our simulation results using synthesized and real-world datasets [8], [32] show that the proposed crossover attention achieves around 1 dB gains on reducing the prediction errors when comparing to the vanilla attention and outperforms recent reference schemes. The novelty of the proposed crossover attention lies on the simple yet effective modification of the vanilla attention mechanism. Extensive ablation simulations verified that performance improvements

are achieved by improving the attention mechanism itself rather than employing more traditional attention layers. This demonstrates the effectiveness of the proposed crossover attention mechanism. To conclude, our contributions are twofold:

- *Enhanced Spatio-Temporal Predictive Learning*: We improved the attention model by learning a regression kernel based on both temporal and spatial similarities, which is crucial for applications like channel prediction, traffic prediction, and more. To the best of our knowledge, our work is the first to introduce a spatial attention layer which captures spatial correlations without relying on convolutional networks. The proposed dual attention layer allows the model to prioritize input sequences that are similar in both dimensions, making it more effective for spatio-temporal predictive learning.
- *Simple but Effective Modification*: The proposed crossover attention is a straightforward modification of the vanilla attention model, making it easy to implement while significantly enhancing performance. This simplicity ensures that it can be readily adopted in various existing frameworks.

II. PRELIMINARIES

A. SPATIO-TEMPORAL PREDICTIVE LEARNING

Let $\mathbf{Z}_{1:T} = \{\mathbf{z}_t\}_{t=1}^T \in \mathbb{R}^{T \times D_z}$ be a multivariate time series, where each vector $\mathbf{z}_t = [z_{t,1}, \dots, z_{t,i}, \dots, z_{t,D_z}] \in \mathbb{R}^{D_z}$ represents the variables observed at time t , and $\mathbf{Z}_{t_1:t_2} \in \mathbb{R}^{(t_2-t_1+1) \times D_z}$ represents all values within the time slice $t \in [t_1, t_2]$. This time series exhibits both temporal and spatial correlations. It may be associated with an independent sequence of covariates, denoted by $\mathbf{X}_{1:T} = \{\mathbf{x}_t\}_{t=1}^T \in \mathbb{R}^{T \times D_x}$, where each vector $\mathbf{x}_t \in \mathbb{R}^{D_x}$ can contains both dynamic and static domain-specific features.

The goal of spatio-temporal predictive learning is to learn the prediction model

$$\mathbf{Z}_{t+1:t+h} \sim p(\mathbf{Z}_{t+1:t+h} | \mathbf{Z}_{t-w+1:t}, \mathbf{X}_{t-w+1:t+h}; \boldsymbol{\theta}), \quad (1)$$

where w represents the maximum length of the moving history window, h represents the prediction horizon, and $\boldsymbol{\theta}$ represents the parameters of the model. This model is then employed to predict the future h steps targets $\mathbf{Z}_{t+1:t+h}$ based on a moving window of past w steps observations and the corresponding covariates. In particular, in the absence of covariates, (1) simplifies to an auto-regressive prediction model. Additionally, one might want to learn a mapping from input features to the parameters of the prediction model as

$$\boldsymbol{\theta} = \Psi(\mathbf{Z}_{1:t}, \mathbf{X}_{1:t+h}; \boldsymbol{\omega}), \quad (2)$$

where $\Psi(\cdot; \boldsymbol{\omega})$ is typically a neural network parameterized by a set of learning parameters $\boldsymbol{\omega}$, such as weights and bias. It is worth noting that $\Psi(\cdot; \boldsymbol{\omega})$ is often used to learn the dependency structure among the time series.

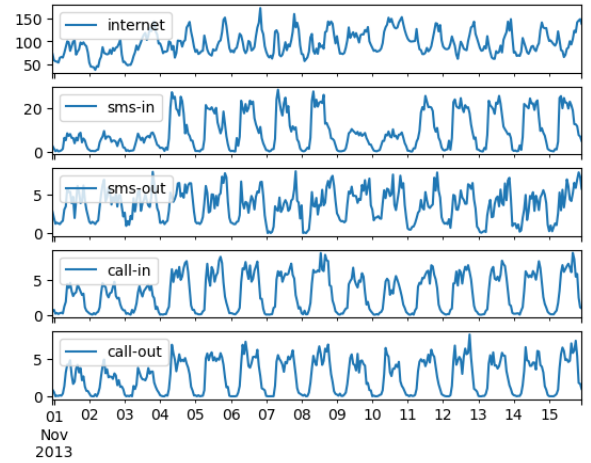


FIGURE 1. Hourly cellular traffic pattern in two weeks of the (50, 50)-th cell in Milan, Italy.

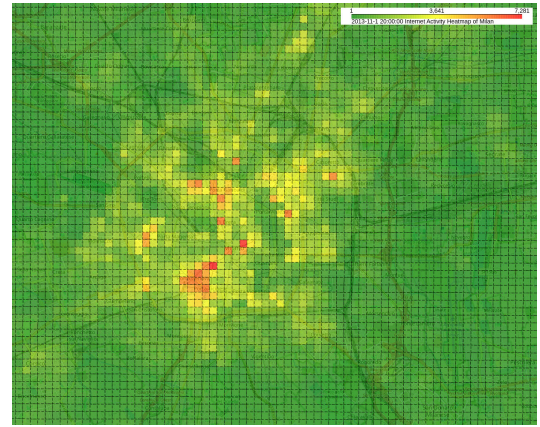


FIGURE 2. Heat map of the internet activities in in Milan, Italy.

B. SPATIO-TEMPORAL DYNAMICS

In practical applications of spatio-temporal predictive learning, the time series frequently exhibits spatial and temporal correlations. Generally, these spatio-temporal dependencies should be leveraged to enhance the prediction accuracy. To illustrate this, we refer to Fig. 1 and Fig. 2, which demonstrate the spatio-temporal effects of cellular traffic data in Milan. This data is sourced from a public dataset released by Italia Telecom [32]. The city is partitioned into a grid of 100×100 cells, each measuring 235×235 square meters. A period of 62 days of communication record details (CDRs) was gathered within this area. The original CDRs, aggregated at 10-minute intervals, were resampled at an hourly interval for this demonstration.

Fig. 1 presents the hourly aggregated traffic data for a specific cell within the city for the first two weeks of November 2013. It is evident that the traffic data follows a distinct seasonal pattern, demonstrating the temporal correlation of the time series. The daily or weekly traffic for a particular cell is correlated and varies in a similar manner. Additionally, Fig. 2 displays a city-wide heatmap of internet activities.

It can be observed that the cellular traffic data collected at neighboring cells varies according to their spatial distribution and traffic data collected within the same zone may exhibit similar variations over time, indicating the spatial correlation of the time series.

C. ATTENTION MECHANISM

The attention mechanism, initially introduced for machine translation in [33], has become a fundamental concept in the deep learning literature. It is designed to capture the long-range dependency structures of the input sequence. The attention mechanism operates as a query-key-value model and typically employs scaled dot-product to calculate the temporal similarities between queries and keys [24]. The outcome is the normalized weighted sum of the training values. The general form of the traditional temporal attention mechanism can be mathematically expressed as

$$\mathbf{A}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \left[\text{softmax} \left(\mathbf{M}_t + \frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{D_k}} \right) \right] \mathbf{V}, \quad (3)$$

where $\mathbf{Q} \in \mathbb{R}^{N \times D_k}$, $\mathbf{K} \in \mathbb{R}^{M \times D_k}$, $\mathbf{V} \in \mathbb{R}^{M \times D_v}$ represent queries and training key-value pairs with lengths of N and M , respectively. In addition, $\mathbf{M}_t \in \mathbb{R}^{N \times M}$ denotes the temporal causality mask for masking out similarities that include future frames, which is achieved by adding $-\infty$ to the corresponding components. The symbol $(\cdot)^T$ represents the matrix transpose, and the $\text{softmax}(\cdot)$ function computes the normalized weights on the last axis of the input tensor. The attention mechanism has been widely used in various spatio-temporal predictive learning tasks, including cellular traffic prediction [1], [2] and MIMO channel prediction [6], [7]. Recent research has adequately confirmed the effectiveness of the attention mechanism in these domains.

III. THE PROPOSED CROSSOVER ATTENTION MECHANISM

Although the attention mechanism has achieved significant success in spatio-temporal predictive learning, it is not designed for efficiently utilizing cross-domain correlations. In this section, we present the proposed direct yet effective variant to augment the capabilities of attention models in spatio-temporal predictive learning.

A. QUERYING BY TEMPORAL CORRELATIONS

The vanilla attention mechanism, as outlined in (3), can be viewed as a realization of the Naradaya-Watson regression model [24]. Let us denote the temporal view of queries, keys, and values as $\mathbf{Q} = \{\mathbf{q}_i\}_{i=1}^N$, $\mathbf{K} = \{\mathbf{k}_i\}_{i=1}^M$, and $\mathbf{V} = \{\mathbf{v}_i\}_{i=1}^M$, where $\mathbf{q}_i \in \mathbb{R}^{1 \times D_k}$, $\mathbf{k}_i \in \mathbb{R}^{1 \times D_k}$, and $\mathbf{v}_i \in \mathbb{R}^{1 \times D_v}$ are the corresponding spatial vectors. Consequently, the Naradaya-Watson regression model can be expressed as

$$\mathbf{a}_i = \sum_{j=1}^M \sigma(\mathbf{q}_i, \mathbf{k}_j) \mathbf{v}_j, \quad \forall i = 1, 2, \dots, N \quad (4)$$

where $\sigma(\cdot)$ denotes a scalar similarity kernel.

In the setting of spatio-temporal predictive learning as outlined in (1), it is appropriate to view key-value pairs as historical covariate-target (features-label) pairs, i.e., $(\mathbf{K}, \mathbf{V}) \triangleq (\mathbf{X}_{1:t}, \mathbf{Z}_{1:t})$. Additionally, queries can be viewed as future h -step covariates, that is, $\mathbf{Q} \triangleq \mathbf{X}_{t+1:t+h}$. The training space contains all observed key-value pairs $\mathcal{D} = \{(\mathbf{k}_i, \mathbf{v}_i)\}_{i=1}^M$, and (4) forecasts the targets by projecting each $\mathbf{q}$ in $\mathcal{D}$ using the similarity kernel $\sigma(\mathbf{q}, \mathbf{k})$.

In the attention mechanism defined in (3), the scaled dot-product similarity between vectors is utilized as the similarity kernel, which is

$$\sigma(\mathbf{q}_i, \mathbf{k}_j) = \text{softmax} \left(\frac{\mathbf{q}_i \mathbf{k}_j^T}{\sqrt{D_k}} \right). \quad (5)$$

This similarity kernel generates the sample cross-covariance matrix $\mathbf{C} = \mathbf{Q}\mathbf{K}^T \in \mathbb{R}^{N \times M}$ between $\mathbf{Q}$ and $\mathbf{K}$, where the (n, m) -th element $C_{n,m} = \text{Cov}(\mathbf{q}_n, \mathbf{k}_m)$ denotes the covariance between the n -th query and the m -th key in $\mathcal{D}$. Consequently, the resulting attention is computed in the temporal domain. In particular, the sample temporal correlation coefficients are explicitly computed in the self-attention $\mathbf{A}(\mathbf{Z}, \mathbf{Z}, \mathbf{Z})$. Therefore, we refer to $\mathbf{A}(\mathbf{Q}, \mathbf{K}, \mathbf{V})$ as the temporal attention *querying by temporal correlations*.

B. QUERYING BY SPATIAL CORRELATIONS

Motivated by the preceding analysis, it is suitable to factorize queries and key-value pairs as the spatial view of $\mathbf{Q} = \{\bar{\mathbf{q}}_i\}_{i=1}^{D_k}$, $\mathbf{K} = \{\bar{\mathbf{k}}_i\}_{i=1}^{D_k}$ and $\mathbf{V} = \{\bar{\mathbf{v}}_i\}_{i=1}^{D_v}$, where $\bar{\mathbf{q}}_i \in \mathbb{R}^{N \times 1}$, $\bar{\mathbf{k}}_i \in \mathbb{R}^{M \times 1}$ and $\bar{\mathbf{v}}_i \in \mathbb{R}^{M \times 1}$ are the corresponding temporal vectors. In this case, the regression model can be adjusted as

$$\mathbf{s}_i = \sum_{j=1}^{D_k} \sigma(\bar{\mathbf{v}}_i, \bar{\mathbf{k}}_j) \bar{\mathbf{q}}_j, \quad \forall i = 1, 2, \dots, D_v \quad (6)$$

with a scalar similarity kernel being the scaled dot-product

$$\sigma(\bar{\mathbf{v}}_i, \bar{\mathbf{k}}_j) = \text{softmax} \left(\frac{\bar{\mathbf{k}}_j^T \bar{\mathbf{v}}_i}{\sqrt{M}} \right). \quad (7)$$

When implementing this regression model as a differentiable neural network layer, it can be expressed as

$$\mathbf{S}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathbf{Q} \left[\text{softmax} \left(\mathbf{M}_s + \frac{\mathbf{K}^T \mathbf{V}}{\sqrt{M}} \right) \right], \quad (8)$$

Now it is evident that the similarity kernel in (8) is computed in the spatial domain, and we employ an optional spatial mask $\mathbf{M}_s$ for masking out unavailable components in the spatial similarity matrix. These unavailable components are typically caused by malfunctioning sensors and can be masked out by adding $-\infty$ before the softmax operation. After the masking, each row of the spatial similarity matrix is normalized using the softmax activation function. This similarity kernel simply generates the sample cross-covariance matrix $\mathbf{G} = \mathbf{K}^T \mathbf{V} \in \mathbb{R}^{D_k \times D_v}$ between $\mathbf{K}$ and $\mathbf{V}$, in which the (k, v) -th element $C_{k,v} = \text{Cov}(\bar{\mathbf{k}}_k, \bar{\mathbf{v}}_v)$ denotes the covariance

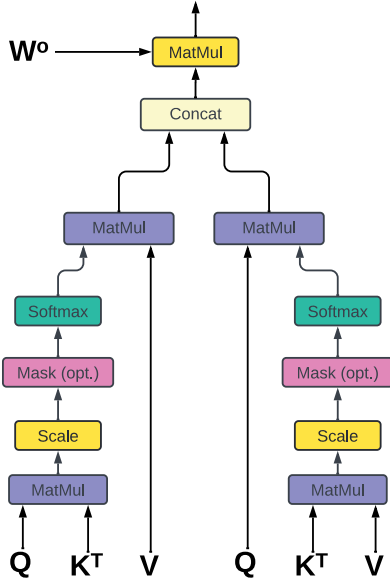


FIGURE 3. Computation graph of the proposed crossover attention mechanism.

between the k -th spatial position of keys and the v -th spatial position of values in $\mathcal{D}$. Specifically, the sample spatial correlation coefficients are explicitly calculated in the self-attention $S(\mathbf{Z}, \mathbf{Z}, \mathbf{Z})$. Therefore, we refer to $S(\mathbf{Q}, \mathbf{K}, \mathbf{V})$ as the spatial attention *querying by spatial correlations*.

C. CROSSOVER ATTENTION

In the implementation of a neural network, the results of querying are determined by the similarity kernel $\sigma(\cdot)$, and the kernel is learned implicitly through the projections $\mathbf{Q} = \mathbf{I}\mathbf{W}^q$, $\mathbf{K} = \mathbf{I}\mathbf{W}^k$ and $\mathbf{V} = \mathbf{I}\mathbf{W}^v$, where $\mathbf{I}$ represents the input to the network layer, and $\mathbf{W}^q$, $\mathbf{W}^k$ and $\mathbf{W}^v$ are learnable projection matrices. To fully utilize the spatio-temporal dependency of the input sequence, the results of querying from the temporal attention and the proposed spatial attention should be integrated. With this consideration in mind, we design the crossover attention as

$$\text{XOA}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = [\mathbf{A}(\mathbf{Q}, \mathbf{K}, \mathbf{V}), \mathbf{S}(\mathbf{Q}, \mathbf{K}, \mathbf{V})] \mathbf{W}^O, \quad (9)$$

where $\mathbf{W}^O \in \mathbb{R}^{2D_v \times D_v}$ is utilized to integrate the attention values computed by temporal and spatial correlations. In general, (9) can be viewed as a component-wise weighted summation of the attention matrices $\mathbf{A}(\mathbf{Q}, \mathbf{K}, \mathbf{V})$ and $\mathbf{S}(\mathbf{Q}, \mathbf{K}, \mathbf{V})$ with the weights controlled by the learnable weight matrix $\mathbf{W}^O$. The structure of the network implementation of the proposed crossover attention is illustrated in Fig. 3. It is important to note that a mask layer is optional before the softmax operation if there are some entries that should be masked out for, e.g., computation purpose. The intuition behind the crossover attention design is rather simple: the integration of the cross-domain attentions could help the neural network to learn a more powerful and expressive regression kernel $\sigma(\cdot)$ which jointly considers the temporal

and spatial dependencies. As will be shown in the simulation results, the introduced crossover attention outperforms the standard attention mechanism, yielding significantly improved prediction results across diverse applications and datasets.

D. COMPLEXITY ANALYSIS

Analyzing the computational complexity of crossover attention is straightforward. The complexity for the crossover attention described in (9) is influenced by both temporal and spatial attentions. For the temporal attention in (3), it involves matrix multiplication between an $N \times D_k$ matrix and a $D_k \times M$ matrix, followed by multiplying the resulting $N \times M$ matrix by an $M \times D_v$ matrix. Thus, the complexity of (3) is generally $\mathcal{O}(NM(D_k + D_v))$. Specifically, for self-attention $\mathbf{A}(\mathbf{Z}_{1:t}, \mathbf{Z}_{1:t}, \mathbf{Z}_{1:t})$, it is $\mathcal{O}(T^2 D_z)$. Similarly, the spatial attention in (8) has a complexity of $\mathcal{O}(D_k D_v (N + M))$, and for self-attention $\mathbf{S}(\mathbf{Z}_{1:t}, \mathbf{Z}_{1:t}, \mathbf{Z}_{1:t})$, it is $\mathcal{O}(TD_z^2)$. Therefore, the total complexity of the proposed crossover attention is $\mathcal{O}(D_k D_v (N + M) + NM(D_k + D_v) + 2ND_v^2)$, and for self-crossover-attention $\text{XOA}(\mathbf{Z}_{1:t}, \mathbf{Z}_{1:t}, \mathbf{Z}_{1:t})$, it is $\mathcal{O}(T^2 D_z + 2TD_z^2)$. For a fixed D_z , it is clear while the computational complexity of the proposed crossover attention is slightly higher than that of traditional attention, it still operates in quadratic time, similar to traditional attention.

Additionally, crossover attention uses the same set of queries, keys, and values for both temporal and spatial attention sub-modules, making it easy to replace the vanilla attention layer with the crossover attention layer. Since the temporal and spatial attentions share the same $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$, the number of learnable parameters in the proposed crossover attention consists only of the weights $\mathbf{W}^q$, $\mathbf{W}^k$, $\mathbf{W}^v$ and $\mathbf{W}^O$. For self-attention, we have $D_k = D_v = D_z$, resulting in $\mathbf{W}^q, \mathbf{W}^k, \mathbf{W}^v \in \mathbb{R}^{D_z \times D_z}$ and $\mathbf{W}^O \in \mathbb{R}^{2D_z \times D_z}$. Consequently, the number of parameters in the proposed crossover attention is $5D_z^2$, whereas the number of parameters in traditional attention is $3D_z^2$.

IV. THE PROPOSED XOATRAN ARCHITECTURE

In this section, we introduce a decoder-only Transformer architecture, namely *XOATrans*, which is constructed based on the proposed crossover attention.

A. MULTI-HEAD CROSSOVER ATTENTION

In practice, transformer models often use multi-head attention mechanism instead of full (single-head) attention [34]. This approach allows the model to jointly attend to information from different representation subspaces at various positions, enhancing the network's expressive power and modeling capabilities. Therefore, we adopt the multi-head pattern for our proposed crossover attention in this paper, and the mathematical model can be expressed as

$$\text{MHXOA}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = [\mathbf{h}_1, \mathbf{h}_2, \dots, \mathbf{h}_H] \mathbf{W}^M, \quad (10)$$

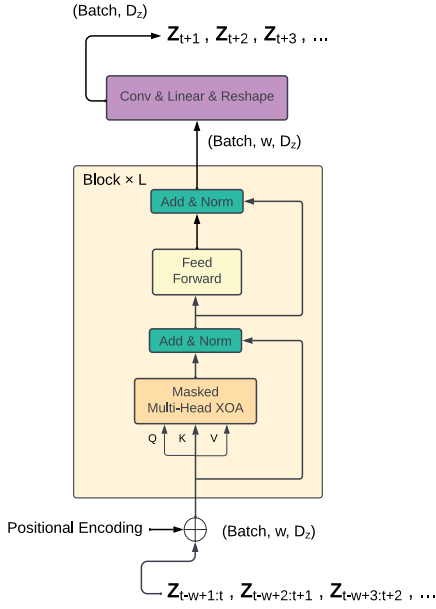


FIGURE 4. Network architecture of XOATran.

where H denotes the total number of heads and each head $\mathbf{h}_i$ is given

$$\mathbf{h}_i = \text{XOA}(\mathbf{Q}\mathbf{W}_i^Q, \mathbf{K}\mathbf{W}_i^K, \mathbf{V}\mathbf{W}_i^V), \quad (11)$$

where $\mathbf{W}_i^Q \in \mathcal{R}^{D_k \times d_k}$, $\mathbf{W}_i^K \in \mathcal{R}^{D_k \times d_k}$, $\mathbf{W}_i^V \in \mathcal{R}^{D_v \times d_v}$ and $\mathbf{W}^M \in \mathcal{R}^{Hd_v \times D_v}$ are parameter matrices for the projections. We denote $d_k = D_k/H$, $d_v = D_v/H$ as the dimensionality of each subspace. Because each head operates on a reduced dimension, the overall computational cost remains similar to that of single-head crossover attention with full dimensionality.

B. MODEL ARCHITECTURE

The Transformer model was initially proposed with an encoder-decoder architecture [34]. However, recent designs of Transformers have favored a decoder-only paradigm, especially for large language models (LLMs) [35], [36]. Following this best practice and simplifying the architecture, we adopt the decoder-only paradigm for our proposed XOATran.

As shown in Fig. 4, the XOATran comprises a positional encoding layer, L decoder blocks, and an output layer for generating the output with expected shape. Unlike RNNs and CNNs, attention layers lack recurrent states and convolutions, potentially losing relative or absolute positional information during forward passes. To maintain the sequential order, we use a fixed positional encoding similar to [34]. The encoded positional information is added to the input, which is then processed by L decoder blocks. Each decoder block includes a masked multi-head crossover attention layer and a feed-forward network. It should be noted that the output of the decoder blocks will retain the same shape as the input. Following the forward pass, the output of the crossover

attention layer will be processed by the Add&Norm operation given by,

$$\text{Add\&Norm}(\mathbf{I}, \mathbf{P}) = \text{Norm}(\mathbf{I} + \mathbf{P}), \quad (12)$$

where $\mathbf{I}$ is the input of this layer, $\mathbf{P}$ is the input of previous layer and $\text{Norm}(\cdot)$ denotes the layer normalization operation [34]. The Add&Norm operation is actually a normalized residual connection which adds the original input to the output of a deeper layer. Residual connections are crucial for the Transformer model architecture. Because they enable effective handling of very deep networks and mitigate the vanishing gradient problem [36]. For the feed-forward network, it can be described as

$$\text{FFN}(\mathbf{I}) = \max(\mathbf{0}, \mathbf{I}\mathbf{W}_1 + \mathbf{B}_1)\mathbf{W}_2 + \mathbf{B}_2. \quad (13)$$

The $\text{FFN}(\cdot)$ layer is essentially a two-layer perceptron with Rectified Linear Unit (ReLU) activation function and $\mathbf{W}_1$, $\mathbf{W}_2$, $\mathbf{B}_1$ and $\mathbf{B}_2$ being the learnable weights and bias with appropriate shapes. Hence, the output of the decoder block can be described as

$$\mathbf{O}_1 = \text{Add\&Norm}(\mathbf{I}, \text{MHXOA}(\mathbf{I}, \mathbf{I}, \mathbf{I})), \quad (14)$$

$$\mathbf{O}_2 = \text{FFN}(\mathbf{O}_1), \quad (15)$$

$$\mathbf{O}_3 = \text{Add\&Norm}(\mathbf{O}_1, \mathbf{O}_2). \quad (16)$$

This process will repeat for L blocks, and the output of the last block will be processed by a CNN layer and/or linear layer to produce the final prediction, depending on the specific application.

C. TRAINING

In this paper, we adopt a supervising learning approach to train the proposed XOATran. Let the final output of the whole network be denoted as

$$\hat{\mathbf{Z}}_{t+1:t+h} = \text{XOATrans}(\mathbf{Z}_{t-w+1:t}), \quad (17)$$

where $\mathbf{Z}_{t-w+1:t}$ is the input sequence with a window length of w , $\hat{\mathbf{Z}}_{t+1:t+h}$ is the h -step prediction of the ground truth $\mathbf{Z}_{t+1:t+h}$. Then, we compute the loss as

$$\mathcal{L}(\mathcal{D}) = \frac{1}{T} \|\text{XOATrans}(\mathbf{Z}_{t-w+1:t}) - \mathbf{Z}_{t+1:t+h}\|^2, \quad (18)$$

$$= \frac{1}{T} \|\hat{\mathbf{Z}}_{t+1:t+h} - \mathbf{Z}_{t+1:t+h}\|^2, \quad (19)$$

where $\mathcal{D} = \{\mathbf{Z}_{t-w+1:t}, \mathbf{Z}_{t+1:t+h}\}_{t=t_0}^{t_0+T}$ is a dataset of T training pairs. The loss function is the mean squared error (MSE) on the training dataset. We employ the PyTorch framework and the Adam stochastic optimizer [37] to implement and train the model.

V. PERFORMANCE EVALUATION

In this section, we present numerical results to verify the effectiveness of the proposed crossover attention in two spatio-temporal predictive learning applications: (i) MIMO channel prediction and (ii) traffic prediction. We also conduct ablation studies by replacing the attention layers inside of

recent developed Transformers with our proposed crossover attention, simplifying the performance comparison.

We adopt several metrics for evaluating the prediction accuracy of each model, including mean absolute error (MAE), mean absolute percentage error (MAPE), mean square error (MSE), normalized root mean square errors (NRMSE), normalized mean squared error (NMSE) and R -squared coefficient of determination (R^2). Note that R^2 provides information about the goodness of a fitting model, and its value normally varies within $[0, 1]$ with 1 indicating the perfect fitting.

In general, all hyper-parameters should be determined based on our available computing resources. While scaling up the model may enhance the network's learning capacity, it can also complicate the training process. Considering these factors, we first establish ranges for all hyper-parameters based on our available computing resources. We then empirically select appropriate values within these ranges, guided by training performance and convergence speed. Notably, for ablation simulations, we follow the same values as specified by the original methods, if applicable.

A. MIMO CHANNEL PREDICTION

One important application of spatio-temporal predictive learning in 5G/6G communications systems is the channel prediction. Predicting CSI from the historical observed CSI could help mitigate the channel aging issue and reduce unnecessary pilot overheads. In this case, we perform simulations for MIMO channel prediction by following a similar problem settings as in [3]. Before showing the simulation results, we will briefly introduce the environment setup.

1) ENVIRONMENT SETUP

We perform simulations based on the downlink of a massive multiuser MISO (MU-MISO) system, where a base station (BS) serves N_u single-antenna users. The system operates in time division duplexing (TDD) mode, and the BS has a limited transmit power P_{tot} , N_t transmit antennas and N_f ($0 < N_f \ll N_t$) RF chains. Initially, the users are randomly located within the coverage area of the BS and are assumed to move with constant velocities. Additionally, the channels between the BS and users are time-varying. On the downlink transmission, the CSI acquisition is accomplished via uplink pilot-assisted channel measurement, and multi-user precoding is then adopted to mitigate inter-user interference. The pilot overhead could be prohibitive with limited RF chains in the system. To tackle this issue, one feasible approach is to estimate only partial CSI at each frame and employ channel prediction to recover the full CSI from the historical incomplete observations [38]. Then, the predicted full CSI will be used for selecting antennas for downlink transmission [3, 39]. Due to the Doppler effect and spatially dependent antenna patterns, real propagation environments often exhibit temporal and spatial correlations [40], which can be leveraged to predict future channel

states. Because only partial CSIs are available in the history windows, this problem is regarded as a partially observable Markov process (POMDP) which is much challenging than the prediction problems with fully observable channel states [14].

Due to the lack of real data set for this use case, we simulate the propagation environment following the same configuration as in [3, 38], which models the channel evolution by a Gaussian-Markov process with the Jakes' model [41], [42], given by

$$\mathbf{h}_{k,t} = \zeta_k \mathbf{h}_{k,t-1} + \sqrt{1 - \zeta_k^2} \mathbf{\Delta}_t, \quad (20)$$

where $\zeta_k \in [0, 1]$ represents the temporal correlation coefficient for user k , and $\mathbf{\Delta}_t \sim \mathcal{CN}(\mathbf{0}, \mathbf{\Sigma})$ is the innovative complex Gaussian i.i.d. in time. The spatially correlated channel vector $\mathbf{\Delta}_t$ follows the Kronecker model [43], and we denote $\alpha \in [0, 1]$ as the spatial correlation coefficient at BS. The value of ζ_k is determined by the maximum Doppler frequency and is inversely proportional to the terminal speed [41], in which $\zeta_k = 1$ represents a static channel and $\zeta_k = 0$ implies that the channel is i.i.d. over time. The fading correlation coefficient can be obtained from Jakes' model given by $\zeta_k = J_0\left(2\pi \frac{v_k f_c T}{C}\right)$, where $J_0(\cdot)$ denotes the zeroth order Bessel function of the first kind, v_k is the speed of user k , C is the speed of light, and T is the frame duration [44]. It should be noted that although an explicit spatio-temporally correlated channel model is adopted here, the algorithm does not have any prior knowledge of the spatio-temporal correlation model. Hence, the prediction algorithm can be applied to any spatio-temporal time series in real-world datasets, as will be verified in Sec. V-B.

In the simulations, the BS is equipped with $N_t = 32$ antennas and $N_f = 16$ RF chains, and $P_{tot} = 100$ watt. The number of users $N_u = 4$, and we adopt a uniform range of speeds from 3.6 km/h to 72 km/h for all users, and the spatial correlation coefficient $\alpha = 0.3$. To train and test the model, the data set is randomly generated with the POMDP introduced in [3, 38]. The process starts by initially selecting an optimal subset of antennas for estimating partial CSI. Subsequently, the full CSI is reconstructed using historical partial estimations. Furthermore, the reconstructed full CSI is used to select antennas for the next frame and the process is repeated. With the POMDP involving, we store at most 1,000,000 time steps of partial CSIs in a ring buffer, and the stored data is partitioned into a training set and a test set with a ratio of 9 : 1. We train the model by using the generated partial CSIs from the training dataset and use the ground-truth CSIs from the same dataset to compute the training losses and update the model's parameters. After updating the parameters at each epoch, we evaluate the model's prediction accuracy using the testing dataset. The number of decoder blocks is $L = 4$, the length of the history window is $w = 24$, and we set the prediction horizon as $h = 1$ since the interactive process only needs to predict one-step targets.

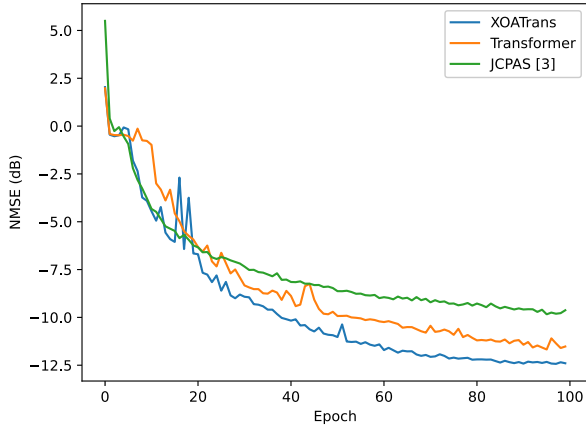


FIGURE 5. NMSE testing results versus training epoch.

To verify the effectiveness of proposed crossover attention mechanism, we perform simulations over the following models,

- *XOATrans*: The proposed crossover attention enabled Transformer introduced in this paper.
- *Transformer*: The widely adopted decoder-only Transformer model with conventional attention mechanism. In particular, this model is constructed by substituting the crossover attention layer of *XOATrans* with the vanilla attention layer.
- *JCPAS*: The joint channel prediction and antenna selection framework introduced in [3], where the probabilistic prediction networks is based on convolutional network with 24 layers and residual connections. It should be noted that JCPAS is designed to output probabilistic results, and we adjust it to output deterministic results for the ease of comparison.

2) NUMERICAL RESULTS

Note that for every epoch we test these models on the same test set, and Figs. 5- 6 illustrates the rolling testing results during the training. Specifically, fig. 5 illustrates the NMSE of prediction results during the training process of the aforementioned models. From Fig. 5, we can observe that all three models are effective in predicting the full CSI from the history of incomplete observations. Additionally, we can see that Transformer model performs better than the JCPAS using CNNs, as the CNNs struggle to capture the long-range temporal dependencies among the input sequence. Moreover, we can conclude from the figure that the proposed *XOATrans* outperforms the two reference models. Specifically, *XOATrans* achieves a gain of about 1 dB and 2.5 dB compared to *Transformer* and *JCPAS*, respectively. This confirms the effectiveness of the proposed crossover attention in capturing both the temporal and spatial structure of the input sequence.

Fig. 6 presents the tested sum-spectral efficiency of the aforementioned models. In this figure, the sum-spectral efficiency is computed using the selected antennas, which are chosen based on the predicted full CSI as in [3]. It should

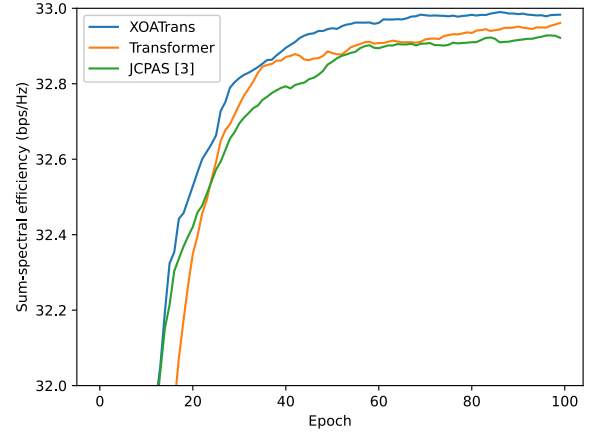


FIGURE 6. Sum-spectral efficiency testing results versus training epoch.

be noted that the sum-spectral efficiency can be maximized by selecting the best subset of antennas, and we adopted the norm-based antenna selection algorithm for this purpose. Therefore, accurate channel prediction will eventually result in higher sum-spectral efficiency. From this perspective, we can conclude from this figure that the proposed *XOATrans* outperforms *Transformer* and *JCPAS*. This is because *XOATrans* not only achieves the lowest NMSE but also preserves the ordering information of the norms of channel vectors, which ultimately helps the algorithm to select a better subset of antennas compared to the other two models. While NMSE directly measures the prediction accuracy, the sum-spectral efficiency of antenna selection is presented as an indirect proxy measurement of the prediction accuracy in Fig. 6. Therefore, the degree of sum-spectral efficiency gain may appear to be less significant than the NMSE gain. However, we can find from the figure that our proposed *XOATrans* converges much faster than *Transformer*, which uses roughly 30% fewer epochs. These findings further verified the effectiveness of the proposed crossover attention.

B. TRAFFIC PREDICTION

In this section, we verify the effectiveness of the proposed crossover attention using two real-world datasets for traffic prediction. Traffic prediction is an important application of spatio-temporal predictive learning. Accurate and timely prediction of the traffics allows more efficient resources allocation and management. Before introducing the results, we will first briefly describe the datasets used in the simulations.

- *Milan dataset* [32]: As shown in Fig. 1, this dataset comprises 62 days of cellular mobile traffic data for the city of Milan, Italy. In this dataset, the entire city area is divided into 100×100 square cells, with cellular traffic data resampled at an hourly granularity. In the experiments, we focus on predicting the number of call-ins for the selected 20×20 cells within this dataset under the same settings in [6].

- *SanDiego* dataset [8]: Compared to the *Milan* dataset, this dataset is significantly larger and is a subset of the LargeST benchmark dataset. It includes road traffic data for over 17,000 road segments and 700 sensors in the area around San Diego, USA. The data is recorded at 5-minute intervals over five years, from 2017 to 2021. In our experiments, we focus on predicting the traffic volumes for all sensors within this dataset.

1) COMPETING ALGORITHMS

We conduct ablation experiments by replacing the attention modules in two existing Transformers: (i) ST-Tran-TTB and (ii) STTN. The details of these Transformers can be found in [6], [8], and [20]. For the reader's convenience, we list below the abbreviations of the competing algorithms or neural networks:

- *HA*: The Historical Average (HA) algorithm, which takes the average of its history as the prediction result.
- *HL*: The Historical Last (HL) algorithm, which simply uses the last observation as the future prediction.
- *ARIMA*: The well-known Autoregressive Integrated Moving Average (ARIMA) algorithm, implemented using the *statsmodels* Python library.
- *LSTM*: The long-short term memory (LSTM) neural network for time-series forecasting [12].
- *ConvLSTM*: The convolutional LSTM proposed for STMTS forecasting in [21].
- *STDenseNet*: STDenseNet [11], a prediction model that learns spatio-temporal dependency structures using densely connected CNNs.
- *DCRNN*: The diffusion convolutional recurrent neural network (DCRNN) proposed in [45].
- *AGCRN*: The adaptive graph convolutional recurrent network (AGCRN) proposed in [46].
- *STGCN*: The spatio-temporal graph convolutional networks proposed in [47].
- *ST-Tran-TTB*: ST-Tran, an encoder-decoder Transformer designed for STMTS forecasting with a temporal transformer block (TTB) [6].
- *STTN*: The spatio-temporal Transformer networks (STTN) for spatio-temporal traffic forecasting [20], which integrate GCNs alongside the attention mechanism.
- *ST-Tran-XOA*: Our modified version of ST-Tran, where the attention layers are replaced by our proposed crossover attention layers.
- *STTN-XOA*: Our modified version of STTN, where the attention layers are replaced by our proposed crossover attention layers.

Note that ST-Tran-XOA and STTN-XOA are tested using the same random seeds, hyper-parameters, instructions, and datasets as described in the original papers. This approach allows us to present a clear and straightforward performance comparison to demonstrate the effectiveness of the proposed crossover attention mechanism.

TABLE 1. Performance comparisons for the Milan dataset.

Model	MAE	NRMSE	R^2
HA	18.7226	0.9687	0.4419
ARIMA	17.1895	0.8813	0.6564
LSTM [12]	13.9438	0.6079	0.7802
STDenseNet [11]	12.3168	0.6442	0.7842
ConvLSTM [21]	11.2308	0.5652	0.8097
ST-Tran-TTB [6]	10.3820	0.5521	0.8187
ST-Tran-XOA	9.9943	0.5508	0.8196

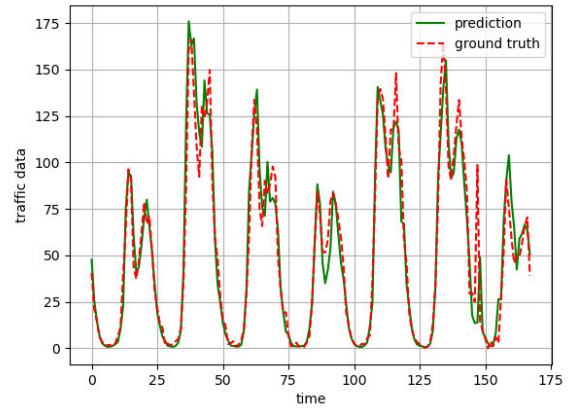


FIGURE 7. The fitness curve of ST-Tran-XOA for the cellular traffic flows in Milan.

2) NUMERICAL RESULT AND DISCUSSION

Table 1 summarizes the prediction performance comparisons of the competing models for the *Milan* dataset. From this table, we can observe that our proposed crossover attention mechanism achieves the best prediction accuracy in terms of MAE, NRMSE, and R^2 . Notably, ST-Tran-XOA attains the highest R^2 score among the seven competing models, indicating that it learns the most fitting model for the spatio-temporal cellular traffic data in the *Milan* dataset. By comparing ST-Tran-XOA with ST-Tran-TTB, we can conclude that the proposed crossover attention mechanism helps the model exploit the spatio-temporal dependencies of the data, resulting in lower prediction errors. In addition to the numerical results presented in Table 1, we also depict the predicted results in Fig. 7 to illustrate the model fitness of ST-Tran-XOA. As shown in the figure, ST-Tran-XOA can accurately and smoothly predict the trends and values of future mobile traffic data for a specific cell within the city area, clearly verifying the effectiveness of the proposed crossover attention mechanism.

In addition to the *Milan* dataset, Table 2 presents the prediction results of the aforementioned models for the *SanDiego* dataset. From this table, we can see that our proposed crossover attention mechanism outperforms the competing models in terms of prediction errors. Specifically, STTN-XOA achieves the lowest prediction errors among the

TABLE 2. Performance comparisons for the SanDiego dataset.

Model	Horizon 3			Horizon 6			Horizon 12			Average		
	MAE	MSE	MAPE	MAE	MSE	MAPE	MAE	MSE	MAPE	MAE	MSE	MAPE
HL	33.61	50.97	20.77%	57.8	84.92	37.73%	101.74	140.14	76.84%	60.79	87.4	41.88%
LSTM [12]	19.03	30.53	11.81%	25.84	40.87	16.44%	37.63	59.07	25.45%	26.44	41.73	17.20%
DCRNN [45]	17.14	27.47	11.12%	20.99	33.29	13.95%	26.99	42.86	18.67%	21.03	33.37	14.13%
AGCRN [46]	15.71	27.85	11.48%	18.06	31.51	13.06%	21.86	39.44	16.52%	18.09	32.01	13.28%
STGCN [47]	17.45	29.99	12.42%	19.55	33.69	13.68%	23.21	41.23	16.32%	19.67	34.14	13.86%
STTN [20]	16.22	26.22	10.63%	18.76	30.98	12.80%	22.62	39.09	16.14%	18.69	31.11	12.82%
STTN-XOA	15.57	25.48	10.01%	17.87	29.78	11.74%	21.76	37.19	14.81%	17.85	29.74	11.72%

seven competing models for prediction horizons from 3 to 12. This demonstrates that the proposed crossover attention mechanism is also effective in multi-step prediction tasks. By comparing STTN-XOA with STTN, we can conclude that the model's capability of capturing spatio-temporal dependencies is significantly enhanced by the proposed crossover attention. These results further validate the effectiveness of the proposed crossover attention.

VI. CONCLUSION

In this paper, we investigated the spatio-temporal predictive learning problem, focusing on predicting spatially and temporally correlated time series such as channel states and traffic flows. To efficiently exploit the spatio-temporal correlations, we designed a simple yet effective crossover attention mechanism to help the network understand the spatio-temporal patterns of input data. Experimental results on two popular applications of channel and network traffic predictions using both synthetic and realistic datasets clearly verified the effectiveness of our proposed crossover attention. Since the proposed crossover attention can be seamlessly integrated into existing models, we believe it offers an attractive method for enhancing the prediction performance of the existing predictive models.

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